Mock for Quiz2

The due date for submitting this assignment has passed.
Due on 2026-04-10, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-10, 20:31 IST

Let AA be a $m \times n$m × n non-zero real matrix, where mm and nn are two non-zero natural numbers strictly greater than 1. Answer the following questions.

2 points
Choose the correct options from the following.

- [x] If $m < n$m < n, then the maximum possible value of rank of AA is mm.
- [ ] If $m < n$m < n, then the minimum possible value of nullity of AA is nn.
- [ ] If $m > n$m > n, then the maximum possible value of rank of AA is mm.
- [x] If $m > n$m > n, then the minimum possible value of nullity of AA is 00.

Yes, the answer is correct.
Score: 2
Accepted Answers:
If $m < n$m < n, then the maximum possible value of rank of AA is mm.
If $m > n$m > n, then the minimum possible value of nullity of AA is 00.

2 points
If AA is the matrix representation of a linear transformation TT from VV to WW, then choose the set of correct options.

- [ ] $dim(V) = m$dim(V) = m.
- [x] TT cannot be zero transformation.
- [x] If $m < n$m < n, then TT cannot be one to one.
- [ ] If $m < n$m < n, then TT cannot be onto.
- [x] If $m \neq n$m = n, then TT cannot be an isomorphism.

Yes, the answer is correct.
Score: 2
Accepted Answers:
TT cannot be zero transformation.
If $m < n$m < n, then TT cannot be one to one.
If $m \neq n$m = n, then TT cannot be an isomorphism.

3 points

Let A_1 A1 and A_2 A2 be two affine spaces defined by the solution space of the system of linear equations $\begin{bmatrix}1&0&1\\0&1&1\\1&-1&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}2\\2\\0\end{bmatrix}$ $\begin{bmatrix}1&0&1&0\\1&0&1&-1\\1&1&0&-1\end{bmatrix}\begin{bmatrix}x1\\x2\\x3\end{bmatrix}=\begin{bmatrix}2\\2\\0\end{bmatrix}$ and $\begin{bmatrix}1&0&1\\2&0&2\\3&0&3\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}2\\4\\6\end{bmatrix}$ $\begin{bmatrix}1&2&3&0\\0&0&1&2\\3&0&0&1\end{bmatrix}\begin{bmatrix}x1\\x2\\x3\end{bmatrix}=\begin{bmatrix}2\\4\\6\end{bmatrix}$ respectively. Let V_1 V1 and V_2 V2 be the vector subspaces of R^3 R3 corresponding to A_1 A1 and A_2 A2.

Choose the correct options.

V_1 V1 has dimension 1, where as the dimension of V_2 V2 is 2.

V_1 V1 has dimension 2, where as the dimension of V_2 V2 is 1.

Both the V_1 V1 and V_2 V2 have dimension 3.

Both the V_1 V1 and V_2 V2 have dimension 1.

Yes, the answer is correct.

Score: 3

Accepted Answers:

V_1 V1 has dimension 1, where as the dimension of V_2 V2 is 2.

Consider the following subsets of R^3 R3, with $\langle\ ,\ \rangle\langle\ ,\ \rangle$ used to denote the standard dot product on R^3 R3, and $\|\ \|\|\ \|$ used to denote the standard length function.

$$S_1 = \{v \in R^3 \mid \langle v, (1,0,0)\rangle = 2,\ \|v\| = 2\}\text{S1} = \{\text{v} \in \text{R3} \mid \langle \text{v}, (1,0,0)\rangle = 2,\ \|\text{v}\| = 2\}$$

$$S_2 = \{u \in R^3 \mid u \in \text{Span}\{(1,0,0)\},\ \|u\| = 2\}\text{S2} = \{\text{u} \in \text{R3} \mid \text{u} \in \text{Span}\{(1,0,0)\},\ \|\text{u}\| = 2\}$$

If nn denotes the number of elements in $S_1 \cap S_2$ S1 ∩ S2, then find the value of nn.

1

Yes, the answer is correct.

Score: 3

Accepted Answers:

(Type: Numeric) 1

3 points

Consider a linear transformation $S: V \to W$S : V → W defined by $S(A) = AB$S(A) = AB, where $B = \begin{bmatrix}1&0\\1&1\end{bmatrix}$B = [1101], VV is the vector space of all upper triangular matrices of order 2 with respect to the usual addition and scalar multiplication in matrices, and WW is the vector space of all matrices of order 2 with respect to the usual addition and scalar multiplication in matrices. Let $\beta = \{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&1\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\0&1\end{bmatrix}\}$ β = {[1000], [0010], [0001]} be an ordered basis of VV and $\gamma = \{\begin{bmatrix}1&0\\0&0\end{bmatrix}, \begin{bmatrix}0&1\\0&0\end{bmatrix}, \begin{bmatrix}0&0\\1&0\end{bmatrix}\begin{bmatrix}0&0\\0&1\end{bmatrix}\}$γ = {[1000], [0010], [0100][0001]} be an ordered basis of WW and CC be the matrix representation of SS with respect to the ordered basis ββ for the domain and γγ for the codomain. Use this information to answer the following sub questions.

2 points
Which of the following option(s) is/are true?
☐ Order of the matrix CC is $3 \times 43 \times 4$.
☑ Order of the matrix CC is $4 \times 34 \times 3$.
☑ CC has no zero row.
☐ CC has only one non-zero column.
Yes, the answer is correct.
Score: 2
Accepted Answers:
Order of the matrix CC is $4 \times 34 \times 3$.
CC has no zero row.
Find the rank of SS.
3
Yes, the answer is correct.
Score: 2
Accepted Answers:
(Type: Numeric) 3
2 points
Find the nullity of the matrix CC.
0
Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 0
1 point

3 points
A norm on a vector space VV is a function

$$\|\cdot\|: V \to R \quad \|\cdot\| : \text{V} \to \text{R}$$
$$v \to \|v\| \quad \text{v} \to \|\text{v}\|$$

satisfying the following conditions:

i) $\|v_1 + v_2\| \leq \|v_1\| + \|v_2\|$ $\|\text{v1} + \text{v2}\| \leq \|\text{v1}\| + \|\text{v2}\|$ for all $v_1, v_2 \in V$ $\text{v1}, \text{v2} \in \text{V}$.
ii) $\|cv\| = |c|\|v\|$ $\|\text{cv}\| = |\text{c}|\|\text{v}\|$ for all $c \in R$ $\text{c} \in \text{R}$ and for all $v \in V$ $\text{v} \in \text{V}$.
iii) $\|v\| \geq 0$ $\|\text{v}\| \geq 0$ for all $v \in V$ $\text{v} \in \text{V}$; $\|v\|\|\text{v}\|$=0 if and only if $v = 0$ $\text{v} = 0$.
Define $V = R^2$ $\text{V} = \text{R2}$ and the function defined as:

$$\|\cdot\| : V \to R \quad \|\cdot\|(\text{x}, \text{y})\| : \text{V} \to \text{R} = \max\{|\text{x}|, |\text{y}|\}$$
$$\|(x, y)\| = max\{|x|, |y|\}$$

Which of the conditions of a norm are satisfied for the above function?
☑ Condition 1.
☑ Condition 2.
☑ Condition 3.
Yes, the answer is correct.
Score: 3

Accepted Answers:
Condition 1.
Condition 2.
Condition 3.

Suppose two publication houses (publication house A and publication house B) have organized a sale of their books. Both of them publish three types of books: novels, poetry collections and collections of short stories. The selling price (in (hundreds) ₹) of these three types of books in publication houses A and B are given as follows:

The publication houses announced that in order to avail these special sale prices, customers have to buy equal number of novels, equal number of poetry collection, and equal number of collection of short stories from each of the publication houses (i.e., if a customer buys xx number of novels, yy number of poetry collections and zz number of collection of short stories from Publication house A; then they have to buy exactly xx number of novels, yy number of poetry collections and zz number of collection of short stories from Publication house B, to avail the benefit of the sale). So there is a map taking the tuple consisting of the number of books of each type bought (Novels, Poetry collections, Collection of short stories) to the prices paid by customers who availed the sale to each of the publication houses, which yields a linear transformation $(T)(T)$ from R^3 R3 to R^2 R2 (where the first and second co-ordinates of the image denotes the prices paid to publication house A and publication house B, respectively).

Answer the following questions using the above information.

2 points
If AA is the matrix representation of TT with respect to the basis $\{(1,0,0),(0,1,0),(0,0,1)\}\{(1,0,0),(0,1,0),(0,0,1)\}$ for R^3 R3 and to the basis $\{(1,0),(0,1)\}\{(1,0),(0,1)\}$ for R^2 R2, then AA is

- $\begin{bmatrix} 2 & 4 \\ 3 & 9 \\ 8 & 16 \end{bmatrix}$
- $\begin{bmatrix} 2 & 3 & 8 \\ 4 & 9 & 16 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Yes, the answer is correct.
Score: 2
Accepted Answers:
$\begin{bmatrix} 2 & 3 & 8 \\ 4 & 9 & 16 \end{bmatrix}$

Find out the rank of TT.
2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2
1 point
If $\{(l, m, n)\}$ {(l, m, n)} is a basis of $ker(T)$ker(T), then find the value of mm if n = 1n = 1.
0
Yes, the answer is correct.
Score: 2
Accepted Answers:
(Type: Numeric) 0
2 points
If AA is the matrix representation of TT with respect to the basis $\{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$ {(1, 0, 1), (0, 1, 1), (1, 1, 0)} for R^3 R3 and to the basis $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)} for R^2 R2, then find out the sum of the elements of the bottom most row of the matrix AA.
58
Yes, the answer is correct.